



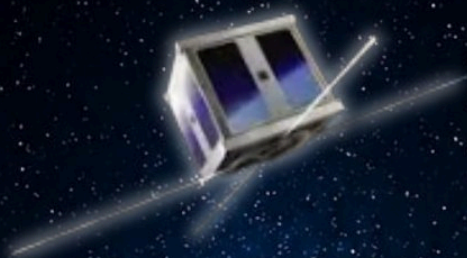
**CHANDIGARH  
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**KALPANA CHAWLA  
CENTRE**

FOR RESEARCH IN  
SPACE SCIENCE &  
TECHNOLOGY



**KCC  
RESPOND  
BASKET 2026**

CU IN SPACE

# K C C R E S P O N D B A S K E T 2 0 2 6

Kalpana Chawla Centre for Research in Space Science & Technology  
Chandigarh University

# DOCUMENT CONTROL

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Document Title</b>    | KCC RESPOND Basket 2026 - Student Project Basket, High-Altitude Balloon Satellite Programme   |
| <b>Document Number</b>   | KCC/BSAT/2026/R1                                                                              |
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| <b>Distribution</b>      | Unrestricted within the institution                                                           |

## Approval Record

| Role                                          | Name | Signature | Date |
|-----------------------------------------------|------|-----------|------|
| Prepared by -<br>Programme Office             |      |           |      |
| Reviewed by -<br>Faculty Review Board         |      |           |      |
| Approved by -<br>Director / Head of<br>Centre |      |           |      |

## Revision History

| Rev. | Date      | Description of Change                                                 |
|------|-----------|-----------------------------------------------------------------------|
| R1   | July 2026 | Initial release - 53 project topics across 10 subsystem work packages |
|      |           |                                                                       |

## For Any Queries Please Contact

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# PREFACE

Hands-on flight projects are the fastest way to turn classroom knowledge into engineering judgement. A high-altitude balloon mission compresses the full life cycle of a space project - requirements, design, fabrication, environmental testing, launch operations, and data analysis - into a single semester, at a cost within reach of a student programme, while exposing hardware to a genuinely space-like environment of -60 °C temperatures, low pressure and high radiation at 25-33 km altitude.

The KCC RESPOND Basket 2026 is the programme's compendium of research and development topics for the coming semester. It defines 53 project topics across ten subsystem work packages, covering the complete flight segment - mission analysis, avionics, structure, mechanisms, science payloads, software and onboard intelligence - together with a full ground segment of antennas, receivers, tracking systems and mission-control software. Each topic states its scope, expected deliverables and linkages to the rest of the mission, so that teams can select work aligned with their interests and the mission's needs.

Projects are deliberately organised by subsystem rather than by academic department: a spacecraft does not care which branch its builders belong to, and students from every discipline are encouraged to take up any topic they can grow into. The five-month plan with its formal review gates - MRR, PDR, CDR, FRR and the Post-Flight Review - mirrors professional aerospace practice and is described in the Mission Timeline section.

We look forward to an inspired semester of building, testing, flying and recovering - and to the data, images and lessons the mission will return.



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**Prof. Eswar Sunkara**  
**Senior Director, KCC**  
**Programme Director**

# GENERAL INSTRUCTIONS

1. The KCC RESPOND BASKET 2026 comprises 53 project topics (45 subsystem projects and 8 ground segment projects) identified for the design, development, testing, launch and recovery of a high-altitude BalloonSat mission reaching 25-33 km altitude, together with a complete student-built ground station.
2. Projects are organised by subsystem work package - Mission Analysis (MSN), Avionics/OBC/EPS (AVI), Structure & Thermal (STR), Mechanisms (MEC), Data Handling (DHA), Science Payloads (PLD/PLC), Software (SWS), AI/ML (AIM) and Ground Segment (GND) - not by academic branch. Students from any discipline may select any project matching their interest and skills.
3. Each project entry comprises a brief write-up (Summary), Scope of the Work, Expected Deliverables, difficulty grade, recommended team size and linkages to other projects of the mission, to help teams select and prepare their work plans.
4. The recommended team size is 2-4 students per project. Each team shall nominate a team lead who attends the weekly all-hands integration meeting.
5. The programme duration is one semester (5 months). Review gates: Mission Requirements Review (Month 1), Preliminary Design Review (Month 2), Critical Design Review (Month 3), Flight Readiness Review (Month 4), Launch + Post-Flight Review and Final Presentations (Month 5). The detailed plan is given in the Mission Timeline section.
6. Mission-level design constraints binding on all teams: total payload mass  $\leq 2$  kg; electronics shall survive -60 °C ambient; a single shared battery bus is preferred and every subsystem shall submit a power budget; telemetry shall use licence-free/amateur bands (e.g. 433 MHz LoRa) within regulatory limits; descent rate at ground level shall be 4-6 m/s; aviation authority clearance (NOTAM) is mandatory before launch.
7. Mission success criteria - Minimum: payload launched, partial telemetry received, payload recovered. Full: continuous telemetry, all sensors logged, live tracking on the mission dashboard, payload recovered intact. Stretch: live images downlinked, all science experiments return usable data, AI/ML models validated on real flight data, results published.
8. Every flying item shall pass environmental acceptance testing (cold soak, drop, vibration, burn-in) on the BSAT-STR-2026-004 test rigs before the Flight Readiness Review.
9. Interface control: pinouts, connectors and the telemetry packet format are controlled documents owned by BSAT-AVI-2026-001 and BSAT-SWS-2026-004; changes are frozen at the Critical Design Review.



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# GROUND STATION & COMMUNICATION SEGMENT

WP GND • 8 PROJECTS

## BSAT-GND-2026-001

### Title of the Project

**High-Gain Yagi Antenna Design & Build**

### Subsystem / Work Package

Ground Segment (GND)

### Area of Work / Key Skills & Tools

Antenna theory, 4NEC2/MMANA-GAL simulation, RF construction, SWR/NanoVNA measurement

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

The Yagi-Uda antenna - a driven dipole plus parasitic reflector and director elements - is how a ground station reaches out 300 km to a milliwatt transmitter: each added director focuses more energy into a narrower beam, trading coverage for gain. This project takes the full RF engineering journey: simulate the design (element lengths and spacing are sensitive at the millimetre level at 433 MHz), build it from aluminum tube or PCB, then measure reality - SWR with a NanoVNA, radiation pattern on an improvised antenna range - and reconcile simulation with measurement. Matching, balun design and weatherproofing complete the craft. Your antenna becomes the station's main aperture, mounted on the BSAT-MEC-2026-003 rotator. Start in simulation software and study why classic published designs choose 7-10 elements.

### Scope of the Work

Design, simulate, build and tune a 433 MHz Yagi-Uda antenna (7-10 elements, ~11-13 dBi) for the main tracking station. Simulate in software, build from aluminum tube/PCB, measure SWR and radiation pattern, weatherproof it and mount on the rotator.

### Expected Deliverables

- Tuned Yagi antenna
- simulation vs measured pattern report
- mounting hardware

### Linkages within the Mission

Avionics, Science Payload

## BSAT-GND-2026-002

### Title of the Project

#### Omnidirectional & Backup Antennas

### Subsystem / Work Package

Ground Segment (GND)

### Area of Work / Key Skills & Tools

Antenna fundamentals, coax/connector work, impedance matching, portable RF design

### Difficulty | Recommended Team Size

Beginner

2 students

### Summary of the Project

Not every antenna should be high-gain: when the balloon is overhead, a Yagi pointing at the horizon misses it entirely, and when the chase team is walking through a field, nobody carries a 2-metre boom. This project builds the supporting antenna cast: a quarter-wave ground-plane omni for reliable all-direction coverage near the station, possibly a collinear for more omni gain, a lightweight wire antenna for the payload itself, and the classic tape-measure Yagi - foldable, nearly indestructible - for handheld direction-finding in the final metres of recovery. Each teaches a different antenna principle, and comparative testing against BSAT-GND-2026-001's Yagi makes the gain/coverage trade-off concrete. Start with the ground-plane omni: one afternoon to build, and it becomes your reference antenna.

### Scope of the Work

Build the secondary receive antennas: quarter-wave ground plane and/or collinear omni for 360° coverage near the station, plus a lightweight airborne antenna for the payload, and a handheld tape-measure Yagi for direction-finding during recovery.

### Expected Deliverables

- Ground-plane omni
- handheld DF Yagi
- airborne antenna
- comparison test report

### Linkages within the Mission

Avionics, Science Payload

## BSAT-GND-2026-003

### Title of the Project

#### SDR Receive Chain & RF Front-End

### Subsystem / Work Package

Ground Segment (GND)

### Area of Work / Key Skills & Tools

Software-defined radio, GNU Radio/SDR software, LNA/filter selection, RF measurements

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

Software-defined radio replaces racks of analog equipment with a \$30 dongle and code - the entire receiver (filtering, demodulation, decoding) becomes software you can inspect and improve. This project builds the station's receive chain: RTL-SDR or similar, a low-noise amplifier mounted at the antenna (the first amplifier sets the noise figure of everything after it - the Friis cascade formula makes this rigorous), a band-pass filter to keep strong local FM/GSM signals from overloading the front end, and the decoding stack for LoRa and other HAB modes. Quantifying sensitivity improvement from each component turns intuition into engineering. Start by receiving known signals with the bare dongle, then add LNA and filter one at a time, measuring the difference.

### Scope of the Work

Build the receiver: RTL-SDR/HackRF setup with low-noise amplifier (LNA) and band-pass filter, gain optimization, receive-chain noise figure estimation, and decoding setup for LoRa/RTTY/APRS. Quantify sensitivity improvement from LNA + filter.

### Expected Deliverables

- Complete SDR receive chain
- sensitivity test report
- decoding software configuration

### Linkages within the Mission

Avionics, Software

## BSAT-GND-2026-004

### Title of the Project

Antenna Rotator Control & Auto-Tracking Software

### Subsystem / Work Package

Ground Segment (GND)

### Area of Work / Key Skills & Tools

Geodesy math (az/el from lat/lon/alt), hamlib/rotctl, control software, Python

### Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

Given the balloon's GPS position and the station's location, where should the antenna point? The math is satellite-tracking geodesy in miniature: convert two lat/lon/altitude coordinates into a local azimuth/elevation pointing vector, accounting for Earth's curvature - at 300 km range and 30 km altitude, the balloon sits barely above the horizon. This project writes the software that closes the loop: parse live telemetry, compute pointing angles, drive the BSAT-MEC-2026-003 rotator via the standard rotctl/hamlib protocol, and - the clever part - dead-reckon during telemetry dropouts by extrapolating the trajectory, so the antenna keeps pointing at where the balloon should be, which is exactly what recovers the link. Start by implementing and unit-testing the az/el math against an online satellite-pass calculator.

## Scope of the Work

The software half of BSAT-MEC-2026-003: compute azimuth/elevation to the balloon from live GPS telemetry, drive the rotator (rotctl/hamlib protocol), predictive pointing during telemetry dropouts, plus manual override joystick mode and a pointing-error dashboard.

## Expected Deliverables

- Auto-tracking software
- dropout prediction mode
- tracking accuracy analysis from flight

## Linkages within the Mission

Software, Mechanisms

# BSAT-GND-2026-005

## Title of the Project

Ground Station Integration & Mission Control Room

## Subsystem / Work Package

Ground Segment (GND)

## Area of Work / Key Skills & Tools

Systems integration, networking, RF station practices, technical documentation

## Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

Antennas, rotator, SDR, decoders, database, dashboard - individually working pieces are not a ground station until someone integrates them, and integration is where projects usually fail. This is the systems-engineering project of the ground segment: define the end-to-end architecture (RF → digital → network → displays), do the unglamorous critical work (cable and connector standards, grounding and lightning protection, UPS backup power, labeling), write the operator manual and checklists, and run full-dress rehearsals tracking a transmitter driven around in a car. On launch day this team operates the station. The mindset is reliability: every single-point failure identified and mitigated. Start by drawing the complete signal-flow diagram with every cable, connector and power source named.



## Scope of the Work

Systems-integration project: assemble the full ground station (antennas → rotator → SDR → decoders → network → dashboard), define the station architecture, cabling/grounding/lightning safety, UPS backup power, operator procedures and station checklists.

## Expected Deliverables

- Fully integrated ground station
- architecture document
- operator manual & checklists

## Linkages within the Mission

All subsystem teams

# BSAT-GND-2026-006

## Title of the Project

**Mobile Chase Ground Station**

## Subsystem / Work Package

Ground Segment (GND)

## Area of Work / Key Skills & Tools

Portable RF systems, power management, vehicle integration, field testing

## Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

The chase team drives toward the predicted landing zone while the balloon is still falling - and they need live telemetry in the car, far from the main station. This project packs a complete receive capability into a grab-and-go kit: magnetic-mount whip antenna on the roof, LoRa receiver or SDR with a laptop or phone, battery power for hours, offline maps, all working while moving. The constraints differ from the fixed station: multipath and obstruction at ground level slash range (quantify it with drive testing), power is finite, and setup time matters - under five minutes from trunk to tracking. Field robustness (vibration, heat, coffee spills) is part of the spec. Start by benchmarking how far you can receive your transmitter from a moving car with the simplest possible setup, then improve the weakest link.

## Scope of the Work

Build a car-portable station for the recovery team: magnetic-mount antenna, LoRa receiver + laptop/phone decoding, battery power, offline mapping. Must set up in under 5 minutes and receive the payload from 20+ km while driving.

## Expected Deliverables

- Complete chase station kit
- drive-test range report
- quick-setup procedure

## Linkages within the Mission

Avionics, Software

**BSAT-GND-2026-007**

### Title of the Project

**RF Link Budget & Propagation Analysis**

### Subsystem / Work Package

Ground Segment (GND)

### Area of Work / Key Skills & Tools

RF propagation theory, link budget math, RSSI data analysis, site surveying

### Difficulty | Recommended Team Size

Medium

2 students

### Summary of the Project

Will the link actually work at 300 km? Link budget analysis answers that question before launch day, in decibels: transmit power, antenna gains, free-space path loss (Friis equation), fade margins for polarization mismatch and payload swing, receiver sensitivity including the LNA's noise figure. This project is the analytical backbone of the entire radio system - it validates every other RF team's choices, selects the ground station site (horizon obstructions matter when the balloon is 1° above the horizon at maximum range), and computes the radio horizon vs altitude. After the flight comes the payoff: comparing predicted against measured RSSI across the entire trajectory, one of the most satisfying validation exercises in the program. Start by building a link budget spreadsheet and sanity-checking it against published LoRa HAB records.

### Scope of the Work

The analytical backbone of the radio system: full link budget (TX power, antenna gains, path loss at 30 km slant range, fade margin), Fresnel zone/horizon analysis, site survey of the ground station location, and validation against measured RSSI from the flight.

### Expected Deliverables

- Link budget document
- station site survey report
- predicted vs measured RSSI study

## Linkages within the Mission

Avionics, Science Payload

**Title of the Project****Multi-Station Network & Community Receivers****Subsystem / Work Package**

Ground Segment (GND)

**Area of Work / Key Skills & Tools**

IoT/networked systems, Raspberry Pi, server backend, distributed systems basics

**Difficulty | Recommended Team Size**

Advanced

3 students

**Summary of the Project**

One receiver station has gaps - terrain blocks low elevations, local noise raises the floor, and a single point of failure risks losing flight data. The amateur HAB community solved this with distributed listener networks, where volunteer stations worldwide forward received packets to a central server. This project builds a miniature version: two or three low-cost autonomous receiver nodes (Raspberry Pi + SDR + omni antenna) deployed at distributed sites - other campuses, students' homes - forwarding packets over the internet to an aggregation server that merges and deduplicates streams. As a stretch goal, differences in packet timing and signal strength across stations enable rough multilateration - estimating the balloon's position without its GPS. Start with one autonomous node that survives a week unattended, then replicate.

**Scope of the Work**

Scale beyond one station: deploy 2-3 low-cost LoRa receiver nodes (Raspberry Pi + RTL-SDR) at distributed sites, aggregate packets over the internet (amateur networks/custom server), improving coverage and enabling rough multilateration experiments.

**Expected Deliverables**

- 3-node receiver network
- aggregation server
- coverage improvement analysis

**Linkages within the Mission**

Software, Avionics, Data Handling

# MISSION ANALYSIS & FLIGHT DYNAMICS

WP MSN • 5 PROJECTS

## BSAT-MSN-2026-001

### Title of the Project

**Flight Trajectory Prediction & Simulation**

### Subsystem / Work Package

Mission Analysis & Flight Dynamics (MSN)

### Area of Work / Key Skills & Tools

MATLAB/Python, atmosphere models (ISA), CUSF/habhub-style predictors, GFS wind data

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

A latex weather balloon does not fly a straight line - it drifts with the wind at every layer of the atmosphere, climbing at roughly 5 m/s until it expands from ~2 m to ~10 m diameter and bursts near 30 km. Predicting this path is a classic aerospace problem combining buoyancy, drag, the standard atmosphere and real forecast wind fields (GFS). Your prediction decides the launch site, the recovery plan and whether the mission is even legal to fly that day (it must not drift into airports or restricted zones). The core challenge is that ascent rate and burst altitude are uncertain, so a single predicted path is never enough - you must understand how errors grow. Start by studying the physics used by the open-source CUSF flight predictor, then build your own simplified model and compare.

### Scope of the Work

Model the complete flight: ascent rate vs free lift, burst altitude prediction, and descent under parachute. Use GFS wind data to predict the flight path and landing zone. Validate the model against the actual flight and publish prediction-vs-actual error analysis.

### Expected Deliverables

- Trajectory simulation tool
- pre-launch prediction report
- post-flight validation report

### Linkages within the Mission

BSAT-DHA-2026-003, Mission Ops

## BSAT-MSN-2026-002

### Title of the Project

#### Parachute & Recovery System Design

### Subsystem / Work Package

Mission Analysis & Flight Dynamics (MSN)

### Area of Work / Key Skills & Tools

Aerodynamics, drag estimation, sewing/fabrication, drop-test experimentation

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

After burst, the payload free-falls in air so thin (1% of sea-level density at 30 km) that the parachute barely works at first - descent starts at 50+ m/s and slows as the air thickens. Your parachute must deliver a safe 4-6 m/s at ground level: too fast destroys the payload, too slow drifts it tens of kilometres further, often into unreachable terrain. This is real aerospace decelerator design: canopy shape selection (hemispherical, cross, toroidal), drag coefficient estimation, and the awkward fact that the parachute must also not tangle with the burst balloon remains above it. The engineering happens in testing - drop tests are your wind tunnel. Start with the drag equation, size a canopy for your payload mass, and design a test campaign that measures real descent rates.

### Scope of the Work

Design, size and fabricate the parachute for a 4-6 m/s landing descent rate. Analyze drag coefficients, perform drop tests from a building/drone, and design the rigging between balloon, parachute and payload train.

### Expected Deliverables

- Flight-ready parachute + rigging
- drop test report
- descent rate analysis

### Linkages within the Mission

BSAT-STR-2026-003, BSAT-MSN-2026-004

## BSAT-MSN-2026-003

### Title of the Project

#### Balloon Sizing, Lift Gas & Burst Analysis

### Subsystem / Work Package

Mission Analysis & Flight Dynamics (MSN)

## Area of Work / Key Skills & Tools

Buoyancy physics, gas laws, safety procedures, trade studies

## Difficulty | Recommended Team Size

Beginner

2 students

### Summary of the Project

Everything about the flight is set at the moment of inflation: how much gas you put in the balloon determines ascent rate, burst altitude and flight duration. Too little lift and the balloon may float for hours and land in another state; too much and it bursts low, wasting the mission. The physics is Archimedes' principle applied to a stretching rubber envelope - free lift, neck lift, gas density (helium vs hydrogen trade-offs including safety), and the manufacturer's burst diameter data. On launch day, your team is the one holding the balloon, so the deliverable is as much procedure as physics: a repeatable way to measure neck lift in the field with wind blowing. Start with the balloon performance calculators used by the HAB community and re-derive their math yourself.

### Scope of the Work

Select balloon size (e.g. 800-1500 g latex), compute required neck lift and gas volume (helium/hydrogen trade study), predict burst altitude, and write the complete launch-day inflation procedure with a lift measurement rig.

### Expected Deliverables

- Balloon selection report
- inflation & neck-lift procedure
- launch-day checklist
- burst altitude prediction

### Linkages within the Mission

BSAT-MSN-2026-001, Launch Ops

## BSAT-MSN-2026-004

### Title of the Project

Payload Train Dynamics & Stability

### Subsystem / Work Package

Mission Analysis & Flight Dynamics (MSN)

## Area of Work / Key Skills & Tools

Rigid body dynamics, IMU data analysis, mechanical design

## Difficulty | Recommended Team Size

Advanced

3 students

## Summary of the Project

Video from most amateur balloon flights is dizzying - the payload spins and swings like a pendulum under a 30-90 m line to the balloon. Understanding why is a genuine dynamics problem: the payload train is a multi-body pendulum excited by wind shear and vortex shedding from the balloon. Reducing the motion matters for camera footage, for the stabilized gimbal team (BSAT-MEC-2026-002), and for scientific sensors that assume a known orientation. Countermeasures include swivels, fins, longer/shorter lines, and mass distribution changes - but you must model and measure before you fix. The IMU flying on the payload gives you real rotation-rate data to validate against. Start by modeling the train as a double pendulum and identifying its natural frequencies.

## Scope of the Work

Study and reduce spin/pendulum motion of the payload during ascent so cameras stay stable. Design the payload train layout (balloon-parachute-radar reflector-payload spacing), swivels and anti-rotation measures. Validate with IMU data from the flight.

## Expected Deliverables

- Payload train design document
- anti-spin hardware
- post-flight IMU stability analysis

## Linkages within the Mission

BSAT-MEC-2026-002, BSAT-AVI-2026-004

## BSAT-MSN-2026-005

### Title of the Project

**Mission Design, Regulations & Launch Operations**

### Subsystem / Work Package

Mission Analysis & Flight Dynamics (MSN)

### Area of Work / Key Skills & Tools

Systems engineering, regulatory research, project management, risk analysis

### Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

Someone must own the mission as a whole - this is the systems engineering role that turns all the subsystem teams' projects into one flight. A high-altitude balloon shares airspace with aircraft, so this project owns the legal side: aviation authority notification (NOTAM), payload mass and line-strength rules, and no-fly-zone analysis. It also owns the master budgets (mass, power, radio bandwidth) that every other team must fit inside, the risk register, and the minute-by-minute launch day plan. It is the least 'build' and most 'engineer' project in the basket, and the best preparation for how real space missions are actually run. Start by reading your country's uncrewed free balloon regulations and writing the top-level mission requirements document.

## Scope of the Work

Own the overall mission: requirements document, mass/power/link budgets roll-up, aviation authority permissions (NOTAM), launch site selection, go/no-go weather criteria, launch-day operations plan and recovery/chase plan.

## Expected Deliverables

- Mission requirements document
- regulatory approvals file
- launch & recovery operations manual
- risk register

## Linkages within the Mission

All teams



# AVIONICS, OBC & ELECTRICAL POWER SYSTEM

WP AVI • 5 PROJECTS

## BSAT-AVI-2026-001

### Title of the Project

**Flight Computer Design & PCB Development**

### Subsystem / Work Package

Avionics, OBC & EPS (AVI)

### Area of Work / Key Skills & Tools

KiCad/Altium, embedded C, PCB assembly & debugging, schematic design

### Difficulty | Recommended Team Size

Advanced

4 students

### Summary of the Project

The flight computer is the heart of the BalloonSat: it reads every sensor, logs the data, feeds the radio and must not crash once during a 3-hour mission where nobody can press reset. At 30 km it faces conditions no classroom project sees: -60 °C ambient, condensation during descent through clouds, battery voltage sagging in the cold, and increased radiation. Designing it means choosing a microcontroller, drawing schematics, laying out a PCB with proper power and connector design, and proving robustness with watchdogs and brown-out handling. The discipline is what matters: version-controlled hardware, bring-up checklists, and a spare unit identical to the flight unit. Start by listing every subsystem that needs power, pins or data - the flight computer is defined by its interfaces.

### Scope of the Work

Design the main flight computer: microcontroller selection (e.g. STM32/ATmega/RP2040), custom PCB with sensor headers, SD logging, watchdog and brown-out protection. Take it from schematic to assembled, environment-tested board.

### Expected Deliverables

- Assembled flight computer PCB (2 units: flight + spare)
- schematics
- test report

### Linkages within the Mission

BSAT-SWS-2026-001, BSAT-AVI-2026-002, BSAT-AVI-2026-004

## BSAT-AVI-2026-002

### Title of the Project

#### Power System for Extreme Cold

### Subsystem / Work Package

Avionics, OBC & EPS (AVI)

### Area of Work / Key Skills & Tools

Power electronics, battery characterization, DC-DC converters, thermal testing

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

Batteries are chemical devices, and chemistry nearly stops at stratospheric temperatures - a battery that delivers full capacity on the bench can die in minutes at -40 °C. This project owns the mission's energy: choosing a chemistry that works cold (lithium primary cells like Li-FeS<sub>2</sub> are the HAB standard), computing the full mission energy budget with margin, designing efficient regulation and distribution, and protecting against shorts in a foam box you cannot open in flight. The proof is experimental: your cold-chamber endurance tests (with the BSAT-PLC-2026-004 cold-test data) decide the flight battery. A power failure kills every other team's project simultaneously, so redundancy thinking is essential. Start by measuring the real current draw of each subsystem prototype, not the datasheet numbers.

### Scope of the Work

Design the power subsystem: energy budget for 3+ hour mission, battery chemistry trade study (Li-primary vs Li-ion at -40 °C), regulators, power distribution, current monitoring, and cold-chamber (dry ice) endurance testing.

### Expected Deliverables

- Power distribution board
- battery cold-test report
- mission power budget spreadsheet

### Linkages within the Mission

BSAT-AVI-2026-001, BSAT-STR-2026-002, All payloads

## BSAT-AVI-2026-003

### Title of the Project

#### Telemetry Radio Link (LoRa / APRS)

### Subsystem / Work Package

Avionics, OBC & EPS (AVI)

## Area of Work / Key Skills & Tools

RF basics, LoRa/packet radio, antenna fundamentals, link budget math

## Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

The radio link is the mission's lifeline - 30+ km of slant range on milliwatts of power, from a transmitter dangling under a balloon to your ground station. LoRa modulation makes this possible for students: its spread-spectrum design decodes signals below the noise floor, so a 25 mW 433 MHz transmitter can reach hundreds of kilometres line-of-sight. Your challenges are the airborne antenna (a quarter-wave wire must be tuned and polarized correctly while swinging), the link budget, and regulatory limits on power and duty cycle. Ground range testing is deceptive - a few km on the ground with obstacles roughly equals hundreds airborne with clear line-of-sight, and you must quantify that. Start with the Friis transmission equation and a pair of LoRa modules on the bench.

### Scope of the Work

Build the airborne transmitter: LoRa 433 MHz (and/or APRS if amateur licensed) sending GPS + sensor packets. Design the airborne antenna, perform range tests (ground-to-ground several km), and characterize the real link budget.

### Expected Deliverables

- Flight transmitter module + airborne antenna
- range test report
- packet format spec (with BSAT-SWS-2026-004)

### Linkages within the Mission

BSAT-GND-2026-001, BSAT-GND-2026-003, BSAT-SWS-2026-004

## BSAT-AVI-2026-004

### Title of the Project

### Flight Sensor Suite Integration

### Subsystem / Work Package

Avionics, OBC & EPS (AVI)

## Area of Work / Key Skills & Tools

Sensor interfacing (I2C/SPI), calibration methods, GPS (airborne dynamic mode), soldering

## Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

A BalloonSat is a flying instrument cluster, and instruments are only as good as their calibration. This project integrates the core sensor stack: GPS (which must support 'airborne' dynamic mode - most consumer GPS chips stop reporting above 18 km due to export rules), barometric pressure (which becomes altitude), internal/external temperature, humidity, UV and a 9-axis IMU. Each sensor has quirks: thermistors self-heat, pressure sensors drift with temperature, magnetometers see the payload's own electronics. Your job is characterizing and correcting these effects so the science-payload and data-handling teams get trustworthy numbers. Start by building one clean I2C/SPI bus plan with the flight computer team, then calibrate sensors one at a time against references.

## Scope of the Work

Select, calibrate and integrate the sensor stack: GPS (high-altitude capable, >18 km), pressure/temperature (internal+external), humidity, UV index, and 9-axis IMU. Deliver calibration curves and an I2C/SPI wiring harness for the flight computer.

## Expected Deliverables

- Calibrated sensor harness
- calibration report per sensor
- sensor datasheet pack

## Linkages within the Mission

BSAT-AVI-2026-001, BSAT-PLD-2026-002, BSAT-SWS-2026-001

## BSAT-AVI-2026-005

### Title of the Project

**Redundant Backup Tracker**

### Subsystem / Work Package

Avionics, OBC & EPS (AVI)

### Area of Work / Key Skills & Tools

Embedded systems, GSM/SIM modules, low-power design

### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Ask any HAB team about their worst moment and it is the silence after the main tracker stops transmitting. Payloads land in trees, ditches and fields where the main antenna is crushed or shadowed - recovery then depends entirely on an independent backup. This project builds a tracker that shares nothing with the main system: its own battery, its own GPS, and a GSM modem that texts the landing position once it descends into cell coverage (GSM works near the ground, not at altitude - understanding why is part of the project). Add a buzzer and strobe for the last 50 metres of the search, often in crops taller than you. The engineering theme is independence and low power. Start by testing GSM cold-start behavior after hours without signal.

### **Scope of the Work**

Build an independent, self-powered backup tracker so the payload is never lost: separate GPS + GSM (SMS position after landing) and/or a second low-power LoRa beacon, plus audio buzzer and LED strobe for final ground search.

### **Expected Deliverables**

- Flight-ready backup tracker (independent battery)
- recovery test report

### **Linkages within the Mission**

Recovery team, BSAT-SWS-2026-005

# STRUCTURE & THERMAL CONTROL

WP STR • 5 PROJECTS

## BSAT-STR-2026-001

### Title of the Project

**Payload Structure & Enclosure Design**

### Subsystem / Work Package

Structure & Thermal (STR)

### Area of Work / Key Skills & Tools

SolidWorks/Fusion 360, 3D printing, foam fabrication, mass budgeting

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

The payload box must survive -60 °C, a violent burst event, a 5 m/s ground impact - and weigh almost nothing, because every gram of structure steals a gram from science payloads. The HAB-standard solution is extruded polystyrene foam (light, insulating, crushable on impact) combined with 3D-printed fittings where precision is needed. This project owns the mechanical architecture: internal layout so cameras see out and antennas radiate down, access for last-minute battery swaps on the launch field, and the master mass budget with authority to push back on overweight subsystems. Packaging nine teams' hardware into ~2 kg is a genuine spacecraft-configuration problem. Start by collecting every subsystem's dimensions and building a CAD mock-up before cutting any foam.

### Scope of the Work

Design the payload box: foam/3D-printed hybrid structure, internal component layout, camera ports, antenna feed-throughs, access hatches. Manage the overall mass budget ( $\leq 2$  kg) and produce full CAD of the assembled payload.

### Expected Deliverables

- Flight enclosure (+1 spare)
- full CAD assembly
- mass budget tracking sheet

### Linkages within the Mission

All flying subsystems

## BSAT-STR-2026-002

### Title of the Project

#### Thermal Design & Cold Environment Testing

### Subsystem / Work Package

Structure & Thermal (STR)

### Area of Work / Key Skills & Tools

Heat transfer, thermal modeling, insulation materials, experimental testing

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

The stratosphere is a thermal paradox: air at  $-60^{\circ}\text{C}$  but too thin to carry heat away effectively, full sun on one face and radiative cold sky on the other. Electronics generate their own heat, so a well-insulated box can stay comfortable inside - but batteries and external sensors need careful placement, and cameras need windows that do not frost over during descent through moist air. This project builds the thermal model of the payload (conduction through foam, internal convection, radiation) and validates it in a dry-ice cold chamber you construct yourself. The interplay with BSAT-AVI-2026-002 is tight: battery placement is a thermal decision. Start with a simple lumped-capacitance model of a foam box with a known heat source, then test it.

### Scope of the Work

Keep electronics alive at  $-60^{\circ}\text{C}$ : insulation design, thermal mass placement, optional heater circuit sizing, thermal model of the payload interior, and validation in a dry-ice cold chamber you build yourself.

### Expected Deliverables

- Thermal analysis report
- DIY cold-test chamber
- validated thermal protection design

### Linkages within the Mission

BSAT-AVI-2026-002, BSAT-PLC-2026-004, BSAT-STR-2026-001

## BSAT-STR-2026-003

### Title of the Project

#### Recovery Hardware & Rigging

### Subsystem / Work Package

Structure & Thermal (STR)

## Area of Work / Key Skills & Tools

Structural design, load testing, materials selection, knots & rigging

## Difficulty | Recommended Team Size

Beginner

2 students

### Summary of the Project

Between the balloon and the payload runs a chain of components that all experience the mission's most violent moment: burst, when drag loads spike and 1+ kg of shredded latex whips around the line. Every attachment point, knot, ring and cord in the load path is this project's responsibility, along with the radar reflector mounting required in many jurisdictions. The rules of thumb are aerospace ones: identify the load path, compute worst-case loads (burst shock, parachute opening, ground impact), then proof-test to 5× - because a single failed knot loses the entire mission. Materials matter too: nylon goes brittle at -60 °C. Start by drawing the complete payload train and labeling the load at every junction.

### Scope of the Work

Design all load-path hardware: attachment points, load rings, shock cord sizing, parachute connection, radar reflector mounting. Proof-load test everything to 5× expected loads including burst shock loading.

### Expected Deliverables

- Flight rigging kit
- proof-load test report
- rigging assembly procedure

### Linkages within the Mission

BSAT-MSN-2026-002, BSAT-STR-2026-001

## BSAT-STR-2026-004

### Title of the Project

### Environmental Test Program & Test Rigs

### Subsystem / Work Package

Structure & Thermal (STR)

## Area of Work / Key Skills & Tools

Test engineering, fixture design, instrumentation, documentation

## Difficulty | Recommended Team Size

Medium

3 students



## Summary of the Project

Real aerospace programs live by 'test like you fly' - every flight component proves itself in ground testing that mimics flight conditions. This project builds the program's test infrastructure and standards: a drop rig for impact, a shake test for launch handling and flight vibration, cold soak procedures, and a burn-in protocol. You define the pass/fail criteria and run every other team's hardware through acceptance testing before it earns a place on the payload - a quality-gate role that teaches test engineering, fixture design and disciplined documentation. The mindset shift is from 'it worked once' to 'it cannot fail'. Start by writing a one-page environmental test standard listing conditions the flight will impose, with numbers.

## Scope of the Work

Build the test infrastructure for the whole program: drop-test rig, vibration/shake test setup, whip-test (for antenna/boom), and a vacuum test using a vacuum chamber if available. Define pass/fail criteria and run acceptance tests for every flying subsystem.

## Expected Deliverables

- Test rigs (drop/vibe)
- environmental test standard document
- acceptance test reports for all subsystems

## Linkages within the Mission

All teams

# BSAT-STR-2026-005

## Title of the Project

Camera Mounts & External Booms

## Subsystem / Work Package

Structure & Thermal (STR)

## Area of Work / Key Skills & Tools

Mechanical design, vibration isolation, 3D printing, optics basics

## Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Cameras are the emotional payload of every BalloonSat - the shot of Earth's curve against black sky is what everyone remembers - but getting it is mechanical engineering. Mounts must hold cameras rigidly through vibration yet isolate them from payload resonances; lenses frost over during descent unless windows are treated or heated; and external sensor probes must sit on booms outside the payload's thermal boundary layer or they measure box temperature, not atmosphere. This project designs those mounts, booms and windows, working closely with the gimbal (BSAT-MEC-2026-002) and sensor (BSAT-AVI-2026-004) teams. Weight is the constant enemy. Start by testing anti-fog approaches in a freezer with a humid transition - the exact condition of descent through cloud.

## Scope of the Work

Design vibration-isolated camera mounts (down-looking + horizon), external sensor booms that place probes outside the payload's thermal boundary layer, and lens anti-fog/anti-frost measures.

## Expected Deliverables

- Camera mount assemblies
- sensor boom
- frost-prevention test results

## Linkages within the Mission

BSAT-MEC-2026-002, BSAT-AVI-2026-004, BSAT-AIM-2026-001

# MECHANISMS & ACTUATION

WP MEC • 5 PROJECTS

## BSAT-MEC-2026-001

### Title of the Project

**Flight Termination / Cut-Down Mechanism**

### Subsystem / Work Package

Mechanisms & Actuation (MEC)

### Area of Work / Key Skills & Tools

Actuator design, embedded control, FMEA, redundancy design

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

A cut-down device separates the payload from the balloon on command - the mission's emergency brake if the flight drifts toward an airport or the border. It is a beautiful mechatronics problem because the requirements conflict: it must fire with 100% certainty when commanded, yet never fire accidentally from vibration, cold, radio noise or a software glitch. The classic actuator is a nichrome hot-wire melting through the nylon line; alternatives are servo releases and pyrotechnic-free spring devices. The real deliverable is the failure-mode analysis (FMEA) and a large test log across temperatures - proving both that it fires and that it doesn't. Start by measuring how nichrome cut time varies from +25 °C to -60 °C.

### Scope of the Work

Design a remotely-commanded and timer-backed cut-down device (nichrome hot-wire or servo release) that separates the payload from the balloon if it drifts toward no-fly zones. Include full failure-mode analysis - it must never trigger accidentally.

### Expected Deliverables

- Flight-ready cut-down unit
- FMEA document
- 20+ successful trigger test log

### Linkages within the Mission

BSAT-AVI-2026-001, BSAT-MSN-2026-005, BSAT-SWS-2026-004

## BSAT-MEC-2026-002

### Title of the Project

#### Actively Stabilized Camera Platform

### Subsystem / Work Package

Mechanisms & Actuation (MEC)

### Area of Work / Key Skills & Tools

Control systems (PID), IMU sensor fusion, servo/gimbal hardware, embedded programming

### Difficulty | Recommended Team Size

Advanced

4 students

### Summary of the Project

The payload spins and swings throughout ascent (see BSAT-MSN-2026-004), which ruins video and pointing-dependent experiments. This project fights back actively: a 1-2 axis gimbal that senses rotation with an IMU and counter-rotates the camera with servo or brushless motors. It is a complete closed-loop control project - sensor fusion to estimate attitude (complementary or Kalman filter), PID tuning, actuator saturation, and the special challenges of -60 °C operation where grease stiffens and servo plastics turn brittle. The final exam is unambiguous: side-by-side stabilized vs unstabilized flight footage. Start with a single-axis tabletop gimbal on a turntable and tune the loop until the horizon stays level.

### Scope of the Work

Build a 1-2 axis camera stabilization system using IMU feedback and servo/brushless gimbal control to counteract payload spin, so video stays horizon-locked. Compare stabilized vs unstabilized footage from the flight.

### Expected Deliverables

- Working gimbal unit
- control loop tuning report
- flight footage comparison

### Linkages within the Mission

BSAT-MSN-2026-004, BSAT-STR-2026-005, BSAT-AVI-2026-001

## BSAT-MEC-2026-003

### Title of the Project

#### Automated Antenna Tracking Rotator

### Subsystem / Work Package

Mechanisms & Actuation (MEC)

## Area of Work / Key Skills & Tools

Motor control, mechanical gearing, inverse pointing math, serial protocols (rotctl)

## Difficulty | Recommended Team Size

Advanced

4 students

### Summary of the Project

A high-gain Yagi antenna hears the balloon 300 km away but only within its narrow beam - so someone must point it, continuously, for three hours, as the balloon crosses the sky. This project builds the machine that does it: a motorized azimuth/elevation rotator with enough torque for a 2 m antenna in wind, position feedback, limit switches, and backlash small enough to keep the beam on target. It pairs with BSAT-GND-2026-004 (the pointing software); your side is motors, gears, structure and the embedded controller speaking the standard rotctl protocol. Commercial rotators cost hundreds of dollars - yours should match them for a fraction. Start by computing required torque and pointing accuracy from the antenna's beamwidth ( $\sim 30^\circ$  for a 7-element Yagi, so  $\pm 5^\circ$  is fine).

### Scope of the Work

Build a motorized azimuth/elevation rotator that automatically points the ground-station Yagi at the balloon using live GPS telemetry. Stepper/DC motors, gearing, limit switches, and rotctl-compatible control software.

### Expected Deliverables

- Working Az/EI rotator with auto-tracking
- pointing accuracy test report

### Linkages within the Mission

BSAT-GND-2026-001, BSAT-GND-2026-004, BSAT-SWS-2026-002

## BSAT-MEC-2026-004

### Title of the Project

Automated Air Sampling Mechanism

### Subsystem / Work Package

Mechanisms & Actuation (MEC)

## Area of Work / Key Skills & Tools

Actuators & valves, pneumatics basics, altitude-triggered control, sealing design

## Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

The chemistry-payload team wants pristine stratospheric air, and this project builds the robot arm of the mission: a sampling mechanism that opens sealed bags or canisters at programmed altitudes (e.g. 10, 20, 30 km), draws in ambient air, and reseals against contamination during descent through lower atmosphere. The mechatronics challenges stack up: valves and servos that actuate at -60 °C, altitude triggering from the flight computer, sample integrity verification, and doing it all within a tight mass budget. Contamination is the silent failure mode - a leaky seal gives the chemistry-payload team ground air and a worthless result, so seal testing is half the project. Start by defining the contamination requirements with the BSAT-PLC-2026-002 team, then prototype one valve-and-bag unit and vacuum-test its seal.

## Scope of the Work

Design a servo-actuated air sampling system: sealed sample bags/canisters with valves that open at programmed altitudes (e.g. 10/20/30 km) and reseal, giving the chemistry-payload team uncontaminated stratospheric air samples.

## Expected Deliverables

- 3-altitude sampling mechanism
- seal integrity test report
- altitude trigger firmware

## Linkages within the Mission

BSAT-PLC-2026-002, BSAT-AVI-2026-001

### BSAT-MEC-2026-005

#### Title of the Project

Smart Recovery Aids Package

#### Subsystem / Work Package

Mechanisms & Actuation (MEC)

#### Area of Work / Key Skills & Tools

Landing detection algorithms, low-power actuation, embedded sensors

#### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Statistically, finding the payload is the hardest part of a balloon mission - GPS gets you to within ~10 metres, but the last 50 metres in a sugarcane field or forest can take hours. This project builds the payload's self-rescue kit: landing detection logic (GPS descent-rate plus accelerometer impact signature), then automatic activation of a high-brightness strobe, a piercing piezo siren, and optionally a servo-raised flag or antenna mast to lift the radio above the crop canopy. The interesting engineering is the landing detector - it must never trigger in flight, yet always trigger on landing, on battery power that must last many hours of searching. Start by recording accelerometer data from drop tests and designing a robust landing signature.

## Scope of the Work

Build the post-landing recovery suite: high-brightness LED strobe, loud piezo siren, servo-deployed flag/antenna raiser (payloads land in crops/trees), all activated automatically by landing detection logic (GPS + accelerometer).

## Expected Deliverables

- Recovery aids module
- landing-detection algorithm validation
- field find-test results

## Linkages within the Mission

BSAT-AVI-2026-005, Recovery team

# DATA HANDLING & ANALYTICS

WP DHA • 5 PROJECTS

## BSAT-DHA-2026-001

### Title of the Project

**Real-Time Telemetry Pipeline & Dashboard**

### Subsystem / Work Package

Data Handling & Analytics (DHA)

### Area of Work / Key Skills & Tools

Python, time-series DB (InfluxDB/TimescaleDB), Grafana/Plotly Dash, data pipelines

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

During flight, telemetry packets arrive every few seconds carrying position, temperatures, battery voltage and science data - and mission control needs to see it as live, trustworthy visualizations, not scrolling text. This project builds the real-time data backbone: ingesting decoded packets, storing them in a time-series database, and serving a dashboard with altitude profiles, sensor graphs, packet-loss statistics and data-quality flags. The engineering challenges are real-world data ones: out-of-order and corrupted packets, duplicates from multiple receiver stations, and graceful behavior during radio dropouts. On launch day your dashboard is the room's single source of truth. Start by defining the packet schema with BSAT-SWS-2026-004 and replaying recorded telemetry from past public HAB flights through your pipeline.

### Scope of the Work

Build the live data pipeline: ingest decoded packets from the ground station, store in a time-series database, and serve a live dashboard with altitude/temperature/battery graphs, packet statistics and data quality indicators during flight.

### Expected Deliverables

- Live telemetry dashboard
- pipeline architecture doc
- launch-day data ops runbook

### Linkages within the Mission

BSAT-SWS-2026-002, BSAT-SWS-2026-003, BSAT-GND-2026-005



## BSAT-DHA-2026-002

### Title of the Project

### Atmospheric Data Analysis & Science Report

### Subsystem / Work Package

Data Handling & Analytics (DHA)

### Area of Work / Key Skills & Tools

Pandas/NumPy, statistics, atmospheric science basics, scientific writing

### Difficulty | Recommended Team Size

Beginner

2 students

### Summary of the Project

A balloon flight is a physics experiment that hands you a beautiful dataset: pressure falling exponentially through two decades, temperature dropping to  $-60^{\circ}\text{C}$  then rising again above the tropopause - the signature of the ozone layer's heating. This project turns raw flight data into science: cleaning and synchronizing multi-sensor time series, reconstructing altitude from both GPS and pressure (they disagree - understanding why is half the lesson), identifying the tropopause, and comparing everything against the ISA standard atmosphere and same-day radiosonde soundings from your national weather service. It is a complete, honest data-analysis project with uncertainty quantification on real, imperfect data. Start by analyzing a published radiosonde dataset so your pipeline is ready before your flight data exists.

### Scope of the Work

Post-flight, analyze the full dataset: temperature/pressure/humidity profiles vs altitude, identify the tropopause, compare against ISA standard atmosphere and same-day radiosonde/weather-model data, quantify sensor errors.

### Expected Deliverables

- Published flight dataset (cleaned)
- atmospheric analysis report
- comparison vs reference data

### Linkages within the Mission

BSAT-PLD-2026-002, BSAT-AVI-2026-004

## BSAT-DHA-2026-003

### Title of the Project

### Landing Zone Prediction Model

### Subsystem / Work Package

Data Handling & Analytics (DHA)

### Area of Work / Key Skills & Tools

Monte Carlo methods, GFS/wind data APIs, geospatial analysis (GeoPandas), ML regression

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

Where will it land? - the question every chase team asks, and pure physics gives an incomplete answer because burst altitude, descent rate and winds are all uncertain. This project treats landing prediction as a probabilistic problem: Monte Carlo simulation sampling over those uncertainties, driven by GFS wind forecasts, producing not a point but a probability ellipse on a map. During flight it gets more interesting - each new telemetry packet lets you re-run the prediction with actual measured winds, shrinking the ellipse in real time for the chase team. Geospatial data science with immediate, visceral validation: you end the day standing where your model said to stand. Start by wiring up GFS data access and simulating descent trajectories from a fixed burst point.

### Scope of the Work

Build a statistical/ML landing-point predictor using wind forecast data (GFS), Monte Carlo simulation over uncertain burst altitude and descent rates. Output a landing probability ellipse map; validate against the real landing spot.

### Expected Deliverables

- Landing prediction tool with probability map
- validation report vs actual flight

### Linkages within the Mission

BSAT-MSN-2026-001, Recovery team

## BSAT-DHA-2026-004

### Title of the Project

Telemetry Anomaly & Health Monitoring

### Subsystem / Work Package

Data Handling & Analytics (DHA)

### Area of Work / Key Skills & Tools

Statistical process control, anomaly detection, alerting systems, Python

### Difficulty | Recommended Team Size

Medium

2 students

## Summary of the Project

Sensors fail quietly: a battery voltage sags slightly faster than it should, an external thermistor reads suspiciously like the internal one, GPS altitude stalls while pressure keeps falling. During a mission nobody has time to stare at raw numbers, so this project builds automated health monitoring: statistical rules and change-point detection over live telemetry streams, raising prioritized alerts in mission control ('battery trend predicts depletion before landing'). Post-flight, the same system generates an anomaly timeline for every subsystem team. It teaches the applied statistics of streaming data - baselines, drift, false-alarm trade-offs - on a system where a missed alarm costs the mission. Start by injecting synthetic faults into replayed telemetry and measuring your detection and false-alarm rates.

## Scope of the Work

Define health rules and statistical anomaly detection for live telemetry (battery droop, temperature runaway, GPS dropout, sensor drift). Alert mission control in real time and auto-generate a post-flight anomaly timeline.

## Expected Deliverables

- Live health-monitoring module with alerts
- post-flight anomaly report

## Linkages within the Mission

BSAT-DHA-2026-001, BSAT-SWS-2026-003

## BSAT-DHA-2026-005

### Title of the Project

Flight Data Portal & Public Outreach Analytics

### Subsystem / Work Package

Data Handling & Analytics (DHA)

### Area of Work / Key Skills & Tools

Web visualization (Plotly/D3/CesiumJS), data publishing, storytelling with data

### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

A balloon mission generates the most shareable content a college program can produce - near-space photos, a 3D flight path, dramatic data - and this project makes it public. Build the program's open-data portal: cleaned datasets with documentation, interactive visualizations (3D flight replay over terrain, altitude-synced photo browser, sensor explorers), and content feeds for the website and social media. There is real data engineering underneath - reproducible processing from raw logs to published data, versioning, metadata standards - plus the storytelling craft of making a temperature profile compelling to a non-scientist. Your work is also the program's face to sponsors and future students. Start by studying how public agencies publish mission data and designing your dataset schema before the flight.

### **Scope of the Work**

Build the program's public face: an open data portal hosting all flight datasets with interactive visualizations (3D flight path, altitude profiles, photo map), plus outreach analytics. Feeds the college website/social media.

### **Expected Deliverables**

- Public flight-data portal
- interactive 3D flight replay
- outreach content pack

### **Linkages within the Mission**

BSAT-DHA-2026-001, BSAT-SWS-2026-003

# SCIENCE PAYLOAD - ATMOSPHERIC PHYSICS

WP PLD • 5 PROJECTS

## BSAT-PLD-2026-001

### Title of the Project

**Cosmic Ray Flux vs Altitude Experiment**

### Subsystem / Work Package

Science Payload - Physics (PLD)

### Area of Work / Key Skills & Tools

Radiation detection, detector calibration, counting statistics, data analysis

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

At ground level, the atmosphere shields us from most cosmic radiation - but a balloon climbs through the shield. Count rates rise with altitude to a maximum around 15-20 km (the Regener-Pfotzer maximum, where primary cosmic rays shower into secondaries faster than the thinning air absorbs them) and then fall again, a discovery that originally came from balloon flights and helped prove cosmic rays were extraterrestrial. This project repeats that classic experiment: fly a Geiger-Müller tube or scintillator, handle counting statistics correctly (counts are Poisson - plan integration times), calibrate on the ground, and shield against the payload's own electronics noise. Your data should clearly show the maximum. Start by characterizing your detector's background rate and dead time in the lab.

### Scope of the Work

Fly a Geiger-Müller counter (or SiPM+scintillator) to measure ionizing radiation count rate from ground to 30+ km, capturing the Regener-Pfotzer maximum (~15-20 km). Calibrate on the ground with known sources; analyze count rate vs altitude/pressure.

### Expected Deliverables

- Flight-ready radiation detector
- calibration report
- cosmic ray flux vs altitude paper

### Linkages within the Mission

BSAT-AVI-2026-001, BSAT-DHA-2026-002

## BSAT-PLD-2026-002

### Title of the Project

#### Stratospheric Atmospheric Profiling

### Subsystem / Work Package

Science Payload - Physics (PLD)

### Area of Work / Key Skills & Tools

Thermodynamics, instrumentation, error analysis, atmospheric physics

### Difficulty | Recommended Team Size

Medium

2 students

### Summary of the Project

Measuring air temperature from a moving balloon sounds trivial and is anything but: a sensor in sunlight reads radiative heating, not air temperature; a sensor with slow response smears the profile across hundreds of metres of ascent; a sensor near the payload measures the box's thermal wake. Professional radiosondes solve these with tiny shielded, aspirated sensors - this project re-derives that engineering: sensor time-constant characterization, radiation shielding, boom placement (with BSAT-STR-2026-005), and altitude reconstruction reconciling GPS with pressure-integrated altitude. Done well, your profile through the tropopause matches the professional soundings; done carelessly, errors of 5-10 °C are typical. Start by measuring your sensor's time constant with a step-change experiment and computing the smearing at 5 m/s ascent.

### Scope of the Work

Design the precision atmospheric measurement experiment: response-time-corrected temperature probes (radiation-shielded, on a boom), pressure-derived vs GPS altitude comparison, lapse rate and tropopause characterization.

### Expected Deliverables

- Shielded probe assembly
- altitude reconstruction methodology
- atmospheric profile analysis

### Linkages within the Mission

BSAT-AVI-2026-004, BSAT-STR-2026-005, BSAT-DHA-2026-002

## BSAT-PLD-2026-003

### Title of the Project

#### UV & Solar Irradiance vs Altitude

### Subsystem / Work Package

Science Payload - Physics (PLD)

### Area of Work / Key Skills & Tools

Radiometry, optical sensor calibration, photometry, ozone physics

### Difficulty | Recommended Team Size

Medium

2 students

### Summary of the Project

The ozone layer absorbs most solar UV-B before it reaches the ground - and your balloon flies up through the absorbing layer, so UV intensity climbs dramatically with altitude, far faster than total sunlight does. This project quantifies that: calibrated UV-A and UV-B sensors plus a total-irradiance reference, flown with known orientation (or corrected using IMU data, since a swinging payload changes the sun angle constantly). The analysis connects measured attenuation to the Beer-Lambert law and the overhead ozone column, tying directly into the ozone payload project (BSAT-PLC-2026-001). Careful radiometry - cosine response, temperature coefficients, cross-sensitivity to visible light - separates good results from noise. Start by cross-calibrating your sensors against a reference on clear days at fixed sun angles.

### Scope of the Work

Measure UV-A/UV-B and total solar irradiance through the flight to quantify ozone-layer attenuation. Calibrate sensors against reference measurements; analyze irradiance vs altitude and solar angle, and estimate ozone column effects.

### Expected Deliverables

- Calibrated UV payload
- irradiance vs altitude analysis
- ozone attenuation report

### Linkages within the Mission

BSAT-AVI-2026-004, BSAT-PLC-2026-001

## BSAT-PLD-2026-004

### Title of the Project

**Geomagnetic Field Mapping Experiment**

### Subsystem / Work Package

Science Payload - Physics (PLD)

### Area of Work / Key Skills & Tools

Magnetometry, sensor fusion, hard/soft iron calibration, IGRF model

### Difficulty | Recommended Team Size

Medium

2 students

## Summary of the Project

Earth's magnetic field is mapped globally by the IGRF model - and your balloon gives you a 30 km vertical transect to test it against. The physics is subtle in a good way: the field changes only ~1% over the altitude range, so this is precision measurement, and the raw magnetometer data is contaminated by the payload itself (hard-iron offsets from magnets and currents, soft-iron distortion from steel) and scrambled by payload rotation. Recovering the true field vector requires calibration ellipsoid fitting and sensor fusion with the IMU - the exact techniques used on real spacecraft magnetometer booms. Agreement with IGRF within a few hundred nanotesla is an excellent result. Start with hard/soft-iron calibration on the ground, rotating the assembled payload, not the bare sensor.

## Scope of the Work

Fly a calibrated 3-axis magnetometer to profile Earth's magnetic field vs altitude and position. Handle payload rotation (fusion with IMU), remove hard/soft-iron effects from the payload itself, compare against IGRF model predictions.

## Expected Deliverables

- Calibrated magnetometer package
- field-vs-IGRF comparison analysis

## Linkages within the Mission

BSAT-AVI-2026-004, BSAT-MSN-2026-004

## BSAT-PLD-2026-005

### Title of the Project

Near-Space Optical Phenomena Study

### Subsystem / Work Package

Science Payload - Physics (PLD)

### Area of Work / Key Skills & Tools

Optics, image analysis, Rayleigh scattering theory, camera calibration

### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Why is the sky blue, and why does it stop being blue? A balloon camera watches Rayleigh scattering disappear in real time: as air thins, the sky overhead darkens toward black while a bright blue rim hugs the horizon. This project does quantitative optics on flight imagery: measuring sky brightness vs altitude from calibrated images (requiring control of camera exposure and response), and the classic exercise of estimating Earth's radius from the horizon's curvature - which demands honest treatment of lens distortion, refraction and the difference between the true and apparent horizon. It is the most accessible physics project and produces the most beautiful figures. Start by calibrating your camera's lens distortion with a checkerboard and locking its exposure settings.



### **Scope of the Work**

Use flight imagery scientifically: measure sky darkening vs altitude (Rayleigh scattering), estimate Earth's radius from horizon curvature photos (correcting lens distortion), and analyze the black-sky/blue-limb transition.

### **Expected Deliverables**

- Camera calibration procedure
- Earth-radius estimate from flight images
- scattering analysis report

### **Linkages within the Mission**

BSAT-STR-2026-005, BSAT-AIM-2026-003

# SCIENCE PAYLOAD - CHEMISTRY & MATERIALS

WP PLC • 5 PROJECTS

## BSAT-PLC-2026-001

### Title of the Project

**Ozone Concentration Profiling**

### Subsystem / Work Package

Science Payload - Chemistry (PLC)

### Area of Work / Key Skills & Tools

Electrochemical sensors, gas calibration, cross-sensitivity analysis, atmospheric chemistry

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

Ozone is a trace gas - parts per million even in the ozone layer - yet it shapes the entire temperature structure of the stratosphere and shields the biosphere from UV. Professional ozonesondes measure it with electrochemical cells flown on weather balloons; this project attempts the same measurement with student-accessible sensors, which is honestly hard: low-cost electrochemical and metal-oxide ozone sensors have strong temperature and pressure dependencies and cross-sensitivities that swamp the signal unless carefully characterized. The project is therefore mostly analytical chemistry on the ground - calibration against known ozone concentrations, compensation model development - with the flight as the final exam. Compare your profile's shape against published ozonesonde data. Start by building an ozone calibration setup with a small UV-lamp ozone generator.

### Scope of the Work

Fly electrochemical/MOS ozone sensors to profile  $O_3$  concentration vs altitude. Major challenge: calibrating low-cost sensors and compensating for temperature/pressure effects. Compare against published ozonesonde profiles.

### Expected Deliverables

- Calibrated ozone payload
- T/P compensation methodology
- ozone profile report

### Linkages within the Mission

BSAT-PLD-2026-003, BSAT-AVI-2026-004

## BSAT-PLC-2026-002

### Title of the Project

#### Stratospheric Air Sampling & Lab Analysis

### Subsystem / Work Package

Science Payload - Chemistry (PLC)

### Area of Work / Key Skills & Tools

Analytical chemistry, GC/GC-MS, sampling protocols, contamination control

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

What is stratospheric air actually made of? Well-mixed gases like CO<sub>2</sub> and CH<sub>4</sub> stay near-constant with altitude, water vapor collapses to a few parts per million, and ozone appears - and with the BSAT-MEC-2026-004 sampling mechanism, you can bring samples home and find out. This project owns everything analytical: a contamination-free sampling protocol (the payload's own outgassing plastics are a contamination source), sample bag/canister material selection, chain-of-custody handling, and lab analysis by GC or GC-MS comparing composition across three altitudes and ground air. The honest challenge is that samples are small and contamination risks are everywhere - rigorous blanks and controls make or break the result. Start by analyzing ground air reproducibly, then vacuum-test the sample containers for storage stability.

### Scope of the Work

Analyze air samples captured by the BSAT-MEC-2026-004 mechanism: design contamination-free sampling protocol, then lab analysis (GC/GC-MS if available) for CO<sub>2</sub>/CH<sub>4</sub>/trace gases; compare compositions at 3 altitudes vs ground air.

### Expected Deliverables

- Sampling protocol document
- lab analysis results
- altitude composition comparison report

### Linkages within the Mission

BSAT-MEC-2026-004

## BSAT-PLC-2026-003

### Title of the Project

#### Aerosol & Particulate Collection Study

### Subsystem / Work Package

Science Payload - Chemistry (PLC)

### Area of Work / Key Skills & Tools

Aerosol science, microscopy (optical/SEM if available), filter chemistry

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

The stratosphere carries a thin haze of aerosols - sulfate particles from volcanic eruptions, meteoric dust, desert dust lofted by storms, and (controversially) viable microbes have all been reported by balloon sampling campaigns. This project builds simple collectors - impaction plates and filters exposed at altitude by a shuttered mechanism - and does the post-flight forensics: optical and (if available) electron microscopy, particle counting and sizing, and elemental analysis to guess at origins. Contamination control is the defining discipline: a fingerprint or launch-field dust ruins everything, so sterile preparation, sealed exposure control samples and blind analysis are essential. Start by exposing collectors at ground level and building your particle-identification skills on known dusts.

### Scope of the Work

Design passive/active particulate collectors (impaction plates, filters) to capture aerosols at altitude; post-flight microscopy and chemical analysis of collected particles; investigate possible stratospheric microbial/dust capture.

### Expected Deliverables

- Collector hardware
- particle analysis report with imagery
- contamination-control assessment

### Linkages within the Mission

BSAT-STR-2026-001, BSAT-MEC-2026-004

## BSAT-PLC-2026-004

### Title of the Project

**Battery Chemistry Performance in Extreme Cold**

### Subsystem / Work Package

Science Payload - Chemistry (PLC)

### Area of Work / Key Skills & Tools

Electrochemistry, battery testing, cold-chamber experimentation, data analysis

### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Every battery datasheet shows capacity at 25 °C; your mission runs at -40 °C and colder, where electrolyte viscosity rises, ion mobility falls, and internal resistance can multiply five-fold. This project is systematic electrochemistry with a mission-critical customer: characterize candidate chemistries (Li-FeS<sub>2</sub> primary, Li-SOCl<sub>2</sub>, Li-ion, NiMH, alkaline) across temperature - capacity retention, voltage under load, internal resistance via pulse testing - in a dry-ice cold chamber. The physical chemistry is rich (Arrhenius behavior of rate processes, Peukert effects) and the output is direct: your data selects the flight battery for BSAT-AVI-2026-002. Lithium primary cells famously dominate HAB flights; your job is to show why, quantitatively. Start by building a constant-current discharge rig with logging, then validate it at room temperature.

## Scope of the Work

Systematic study of battery chemistries (Li-SOCl<sub>2</sub>, Li-FeS<sub>2</sub>/Energizer Lithium, Li-ion, NiMH, alkaline) at -60 to +25 °C: capacity, voltage sag, internal resistance vs temperature. Directly drives BSAT-AVI-2026-002's battery selection.

## Expected Deliverables

- Battery performance database vs temperature
- recommendation report for flight battery

## Linkages within the Mission

BSAT-AVI-2026-002, BSAT-STR-2026-002

## BSAT-PLC-2026-005

### Title of the Project

**Materials Degradation in Near-Space Conditions**

### Subsystem / Work Package

Science Payload - Chemistry (PLC)

### Area of Work / Key Skills & Tools

Polymer chemistry, materials characterization (FTIR/UV-Vis), experimental design

### Difficulty | Recommended Team Size

Medium

2 students

## Summary of the Project

Materials that behave at room temperature turn treacherous in near-space: PLA prints go glassy and crack at -60 °C, adhesives embrittle, plasticizers stiffen, and unfiltered UV at altitude is harsher than anything at ground level. This project flies a coupon board - small standardized samples of polymers, adhesives, coatings and 3D-print filaments - exposed to the full flight environment, with an identical control set kept on the ground. Pre- and post-flight characterization (FTIR for chemical changes, tensile or flex tests for mechanical ones, colorimetry for UV damage) turns anecdotes into a materials selection guide future missions will actually use. One flight is a short exposure, so sensitivity of your characterization methods matters. Start by testing your candidate materials in a freezer and under a UV lamp to rehearse the analysis.

## Scope of the Work

Fly a coupon board of materials (polymers, adhesives, coatings, 3D-print filaments) exposed to UV + cold + low pressure; pre/post characterization (FTIR, tensile, visual) to build a materials selection guide for future missions.

## Expected Deliverables

- Exposed coupon experiment
- pre/post analysis report
- materials selection guide

## Linkages within the Mission

BSAT-STR-2026-001, BSAT-PLD-2026-003

# FLIGHT & GROUND SOFTWARE

WP SWS • 5 PROJECTS

## BSAT-SWS-2026-001

### Title of the Project

**Flight Software & Onboard Data System**

### Subsystem / Work Package

Software Systems (SWS)

### Area of Work / Key Skills & Tools

Embedded C/C++, RTOS or superloop design, state machines, fault-tolerant design, Git

### Difficulty | Recommended Team Size

Advanced

4 students

### Summary of the Project

Flight software is where computer science meets consequences: once released, no one can restart the program for three hours, and every bug flies with you. The architecture is a mission state machine - pre-launch, ascent, burst, descent, landed - because correct behavior differs by phase (transmit fast during descent for recovery; save power after landing). Around it: sensor polling with per-device fault isolation (one frozen I2C sensor must not hang the loop), dual-redundant logging, watchdog-supervised recovery from crashes, and defensive handling of every 'impossible' input. The testing culture is the real lesson - fault injection, 24-hour burn-ins, and code review as if lives depended on it. Start by writing the state machine and its transition rules on paper before any code, then build the fault-injection harness alongside the firmware.

### Scope of the Work

Write the mission-critical flight firmware: state machine (pre-launch/ascent/burst/descent/landed), sensor polling, dual logging (SD + flash), telemetry packet generation, watchdog recovery, safe-mode behavior. Extensive fault-injection testing.

### Expected Deliverables

- Flight firmware (tested via fault injection)
- software design document
- 24-hour burn-in test log

### Linkages within the Mission

BSAT-AVI-2026-001, BSAT-AVI-2026-004, BSAT-SWS-2026-004

## BSAT-SWS-2026-002

### Title of the Project

**Ground Station Receiver Software**

### Subsystem / Work Package

Software Systems (SWS)

### Area of Work / Key Skills & Tools

Python/C++, serial & SDR interfaces, packet parsing, databases, networking

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

Radio hands you a stream of bytes that are sometimes wrong, sometimes duplicated, sometimes missing - turning that into clean mission data is the ground software's job. This project builds the receive-side pipeline: interfacing to LoRa modules and SDR decoders, validating packet CRCs, parsing the binary format into structured records, archiving everything (raw and decoded - you will re-analyze later), and serving a network API that mission control and the dashboard consume. The elegant subproblem is multi-station merging: several receivers hear the same flight with different gaps, and combining their logs recovers packets no single station caught. Reliability engineering matters - this software must run for hours unattended on launch day. Start by defining the packet spec with BSAT-SWS-2026-004 and testing with deliberately corrupted packets.

### Scope of the Work

Build the receive chain software: interface to LoRa receiver/SDR, packet decoding, CRC validation, decoded-packet database logging, and a network API that feeds mission control and the dashboard. Include multi-station packet merging.

### Expected Deliverables

- Ground decoding software suite
- multi-station merge tool
- decoded packet archive from flight

### Linkages within the Mission

BSAT-GND-2026-003, BSAT-GND-2026-005, BSAT-DHA-2026-001

## BSAT-SWS-2026-003

### Title of the Project

**Mission Control Web Application**

### Subsystem / Work Package

Software Systems (SWS)



## Area of Work / Key Skills & Tools

Full-stack web (React/Vue + backend), WebSockets, mapping APIs (Leaflet), mobile-responsive design

## Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

Mission control is a software product with the hardest kind of user: stressed people making time-critical decisions. This project builds the web application the whole program watches on launch day - live map with the balloon's track and predicted landing zone, real-time telemetry graphs, a go/no-go checklist tool for launch procedures, an event log, and a mobile view for the chase team navigating farm roads toward the landing point. The technical stack is modern full-web (WebSockets for live data, mapping libraries, responsive design), but the differentiating skill is UX under pressure: information hierarchy, glanceability, and graceful behavior when the data feed drops. Start by storyboarding launch day minute-by-minute and designing screens for each phase before writing code.

### Scope of the Work

Build the mission control system: live map with balloon position + predicted path, telemetry graphs, go/no-go checklist module, event log, chase-team mobile view with navigation to live/predicted landing point.

### Expected Deliverables

- Deployed mission control web app
- chase-team mobile view
- launch-day usage report

### Linkages within the Mission

BSAT-SWS-2026-002, BSAT-DHA-2026-001, BSAT-DHA-2026-003

## BSAT-SWS-2026-004

### Title of the Project

### Telemetry Protocol & Error Correction Design

### Subsystem / Work Package

Software Systems (SWS)

### Area of Work / Key Skills & Tools

Communication protocols, coding theory (FEC), binary serialization, simulation testing

## Difficulty | Recommended Team Size

Advanced

3 students

## Summary of the Project

Every byte transmitted from the balloon costs airspace, battery and regulatory duty-cycle allowance - so the telemetry protocol is an exercise in engineering under scarcity. This project designs the mission's language: a compact binary packet format (bit-packed fields, not JSON), CRC integrity checks, forward error correction (Reed-Solomon, as used by deep-space missions) so corrupted packets can be repaired rather than discarded, and priority scheduling that guarantees position packets even when science data queues up. The theory is real computer science - information theory, coding theory, serialization - validated by a packet-loss simulator that injects errors and measures recovered throughput. Your specification document becomes law for three other teams. Start by computing your true channel budget: bytes per minute within duty-cycle limits.

## Scope of the Work

Design the radio protocol: compact binary packet format, CRC + forward error correction (Reed-Solomon/convolutional), priority scheduling (position > science data), retransmission strategy, bandwidth budgeting. Test under simulated packet loss.

## Expected Deliverables

- Protocol specification document
- encoder/decoder libraries (flight + ground)
- loss-tolerance test report

## Linkages within the Mission

BSAT-AVI-2026-003, BSAT-SWS-2026-001, BSAT-SWS-2026-002

## BSAT-SWS-2026-005

### Title of the Project

**Recovery & Chase Coordination App**

### Subsystem / Work Package

Software Systems (SWS)

### Area of Work / Key Skills & Tools

Mobile/PWA development, offline-first design, geolocation, SMS gateway integration

### Difficulty | Recommended Team Size

Beginner

2 students

## Summary of the Project

Recovery is a software problem disguised as a hike: the chase team is in a car with patchy internet, trying to converge on a moving prediction, then searching a field for a white box. This project builds their toolkit as an offline-first mobile app/PWA: cached maps for rural dead zones, live and last-known balloon positions, the BSAT-DHA-2026-003 landing ellipse, team member location sharing, navigation to the target, and a direction-finding mode that visualizes LoRa signal strength as you walk. It also integrates the BSAT-AVI-2026-005 backup tracker's SMS positions - parsing texts into map pins automatically. Offline-first architecture is the core discipline: every feature must degrade gracefully without connectivity. Start by mapping the recovery workflow end-to-end, then build the offline map caching first - everything else layers on it.

## Scope of the Work

Build the recovery toolkit: mobile app/PWA showing last-known + predicted position, offline maps for rural areas, direction-finding helper (signal strength display), team location sharing, and integration of the GSM backup tracker's SMS positions.

## Expected Deliverables

- Chase app with offline maps
- SMS tracker integration
- field trial report

## Linkages within the Mission

BSAT-AVI-2026-005, BSAT-DHA-2026-003

# AI / ML & ONBOARD INTELLIGENCE

WP AIM • 5 PROJECTS

## BSAT-AIM-2026-001

### Title of the Project

**Onboard Edge-AI Image Classification**

### Subsystem / Work Package

AI/ML Systems (AIM)

### Area of Work / Key Skills & Tools

TensorFlow Lite/ONNX, model quantization, embedded Linux, power-constrained inference

### Difficulty | Recommended Team Size

Advanced

4 students

### Summary of the Project

The radio downlink is a soda straw - kilobytes per minute - while the onboard camera fills gigabytes. The modern answer is edge AI: run a lightweight neural network on the payload itself to decide which images matter. This project deploys a quantized CNN on a Raspberry Pi (or similar) inside the payload to classify frames - cloud structures, clear ground views, horizon shots - and downlink only tiny thumbnails of the best, effectively giving mission control eyes during flight. The constraints are the interesting part: a few watts of power budget shared with heaters, cold effects on the compute module, and model compression (TFLite/ONNX quantization) to hit inference-per-second targets. Start by collecting training data from public HAB flight footage and benchmarking model families on your target hardware.

### Scope of the Work

Deploy a lightweight CNN on a Raspberry Pi/edge device in the payload to classify images in-flight (cloud types, ground vs sky, interesting scenes) and prioritize which thumbnails to downlink over the slow radio channel.

### Expected Deliverables

- Onboard inference system
- model accuracy report
- smart-downlink flight demonstration

### Linkages within the Mission

BSAT-STR-2026-005, BSAT-AVI-2026-003, BSAT-SWS-2026-004

## BSAT-AIM-2026-002

### Title of the Project

**ML-Enhanced Flight Path Prediction**

### Subsystem / Work Package

AI/ML Systems (AIM)

### Area of Work / Key Skills & Tools

scikit-learn/XGBoost, time-series ML, feature engineering with weather data, model evaluation

### Difficulty | Recommended Team Size

Advanced

3 students

### Summary of the Project

Physics-based flight predictors (BSAT-MSN-2026-001) rest on simplifying assumptions - constant drag coefficient, manufacturer burst diameter, smooth wind interpolation - and systematically miss in ways ML can learn. This project mines the thousands of publicly logged amateur HAB flights plus archived weather data to train models that predict burst altitude and refine landing zones, then benchmarks honestly against the physics baseline on held-out flights. It is a disciplined applied-ML project: feature engineering from weather fields, leakage-free validation (you cannot train on flights from the day you are predicting), uncertainty quantification, and the humility to report if physics wins. The deliverable includes a live prediction API for mission control. Start by scraping and cleaning historical flight logs - the dataset is the project's real asset.

### Scope of the Work

Train ML models (gradient boosting/LSTM) on historical HAB flight data + weather archives to predict burst altitude and improve landing prediction beyond physics-only models. Quantify improvement vs the BSAT-MSN-2026-001/BSAT-DHA-2026-003 baseline.

### Expected Deliverables

- Trained prediction models
- benchmark vs physics baseline
- prediction API for mission control

### Linkages within the Mission

BSAT-MSN-2026-001, BSAT-DHA-2026-003

## BSAT-AIM-2026-003

### Title of the Project

**Computer Vision Post-Flight Analysis**

### Subsystem / Work Package

AI/ML Systems (AIM)

### Area of Work / Key Skills & Tools

OpenCV, feature detection, image stitching, georegistration, Python

### Difficulty | Recommended Team Size

Medium

3 students

### Summary of the Project

A flight returns thousands of images tied to a GPS track - a natural computer-vision dataset with ground truth attached. This project builds the post-flight CV toolkit: automatic horizon detection (a surprisingly deep problem when the horizon is a fuzzy atmospheric gradient), altitude estimation from image features validated against GPS, stitching rotating-payload frames into panoramas, and georegistration - matching flight photos onto satellite imagery to determine exactly what the camera saw. Each task is classical CV (edge detection, feature matching, homographies) with modern options (learned features) available for comparison. The altitude-from-images study is genuinely novel territory for a student project. Start with OpenCV horizon detection on public HAB footage so the pipeline is ready before your flight.

### Scope of the Work

Apply CV to flight imagery: automatic horizon detection, altitude estimation from image features, image stitching into panoramas, GPS-photo georegistration onto satellite maps, and automatic best-photo selection.

### Expected Deliverables

- CV analysis toolkit
- georegistered photo map
- altitude-from-image validation study

### Linkages within the Mission

BSAT-PLD-2026-005, BSAT-STR-2026-005

## BSAT-AIM-2026-004

### Title of the Project

**ML Anomaly Detection on Telemetry Streams**

### Subsystem / Work Package

AI/ML Systems (AIM)

### Area of Work / Key Skills & Tools

Unsupervised ML, autoencoders, streaming inference, synthetic data generation

### Difficulty | Recommended Team Size

Medium

3 students

## Summary of the Project

Rule-based health monitoring (BSAT-DHA-2026-004) catches failures someone anticipated; ML anomaly detection catches the ones nobody did. This project trains unsupervised models - autoencoders that learn to reconstruct 'normal' telemetry and flag what they cannot, isolation forests for point anomalies - on simulated and replayed flight data, then runs live during the mission alongside the rule-based system in a head-to-head comparison. The honest challenges are the ML-engineering ones: almost no labeled anomalies exist, so you must generate realistic synthetic faults; a balloon flight is inherently non-stationary (everything changes with altitude), so 'normal' is a moving target; and false alarms erode trust fast. Producing a labeled anomaly dataset for future teams is a lasting contribution. Start by building the synthetic fault injector - it is your training-data factory.

## Scope of the Work

Go beyond rule-based monitoring: train autoencoder/isolation-forest models on simulated + real telemetry to detect subtle anomalies (slow sensor drift, unusual dynamics). Run live during flight alongside BSAT-DHA-2026-004 and compare detection performance.

## Expected Deliverables

- ML anomaly detection service
- rule-based vs ML comparison study
- labeled anomaly dataset

## Linkages within the Mission

BSAT-DHA-2026-004, BSAT-DHA-2026-001

## BSAT-AIM-2026-005

### Title of the Project

AI-Assisted Signal Decoding & Denoising

### Subsystem / Work Package

AI/ML Systems (AIM)

### Area of Work / Key Skills & Tools

DSP fundamentals, deep learning for signals (CNN/RNN on IQ data), SDR recording, PyTorch

### Difficulty | Recommended Team Size

Advanced

3 students

## Summary of the Project

When the balloon is 300 km out and the signal fades into noise, classical decoders simply reject corrupted packets - but the information is not entirely gone, and machine learning can sometimes claw it back. This research-flavored project records raw IQ samples (the complex baseband signal) at the SDR during flight, augments clean recordings with realistic synthetic noise and fading to build training data, and trains neural networks (CNNs/RNNs on IQ sequences) to denoise signals or classify symbols below the threshold where conventional demodulation fails. Post-flight, the model attempts recovery of the real packets lost at maximum range - a measurable, honest benchmark. It requires learning DSP fundamentals alongside deep learning, a genuinely valuable combination. Start by recording your own transmitter at controlled SNR levels and reproducing a classical demodulator in Python first.

## Scope of the Work

Research project: use ML to recover weak/corrupted radio packets that classical decoding rejects - train denoising models on recorded IQ samples with synthetic noise, and attempt recovery of packets lost during the real flight.

## Expected Deliverables

- IQ dataset with noise augmentation
- trained denoising model
- packet recovery rate study

## Linkages within the Mission

BSAT-GND-2026-003, BSAT-SWS-2026-004



# MISSION TIMELINE - ONE SEMESTER (5 MONTHS)

## Month 1 - Research, Training & Requirements

**All-Teams Milestones:** Kick-off; teams formed; literature review of past BalloonSat/HAB missions; training workshops (soldering, CAD, Git, radio basics); each team writes requirements + concept document; regulatory process started (BSAT-MSN-2026-005).

**Flight Segment Focus:** Define mission requirements; mass/power/data budgets drafted; sensor & component selection; order long-lead parts.

**Ground Segment Focus:** Radio band selection & licensing check; antenna and SDR component selection; station site survey; link budget first draft (BSAT-GND-2026-007).

**Gate 1: Mission Requirements Review (MRR) - end of Month 1**

## Month 2 - Preliminary Design & Breadboarding

**All-Teams Milestones:** All subsystems breadboarded/prototyped; interface control document v1 (pins, connectors, packet fields, mounting points) agreed between all teams; test standards defined (BSAT-STR-2026-004).

**Flight Segment Focus:** Flight computer breadboard running sensors; power system prototype; enclosure CAD v1; parachute sizing done; science payloads proof-of-concept.

**Ground Segment Focus:** First antennas built and SWR-tuned; SDR receiving test transmissions; decoding software receiving dummy packets end-to-end.

**Gate 2: Preliminary Design Review (PDR) - end of Month 2**

## Month 3 - Detailed Design, Fabrication & Subsystem Testing

**All-Teams Milestones:** PCBs fabricated and assembled; enclosures printed/cut; flight software feature-complete; all science experiments calibrated in lab; first full-stack radio test (flight TX → ground RX → dashboard).

**Flight Segment Focus:** Flight PCB bring-up; sensor calibration campaigns; parachute drop tests; cut-down mechanism trigger testing; gimbal control loop tuning.

**Ground Segment Focus:** Rotator assembled; auto-tracking software driving rotator from simulated GPS; mission control web app live with test data; chase station kit assembled.

**Gate 3: Critical Design Review (CDR) - end of Month 3**

## Month 4 - Integration, Environmental Testing & Flight Readiness

**All-Teams Milestones:** Full payload integrated; mass budget locked; environmental test campaign: cold soak (dry ice, -60 °C), drop tests, vibration, 24 h burn-in; ground rehearsal with car/drone range tests (10+ km); regulatory clearance (NOTAM) confirmed.

**Flight Segment Focus:** Fix integration issues; fault-injection testing of flight software; backup tracker verified independently; final flight readiness of all science payloads.

**Ground Segment Focus:** Full ground station integrated & rehearsed; multi-station nodes deployed; tracking rehearsal against a moving transmitter; operator checklists finalized.

**Gate 4: Flight Readiness Review (FRR) - end of Month 4**

## Month 5 - Launch Campaign, Flight, Data Analysis & Showcase

**All-Teams Milestones:** LAUNCH WINDOW (first half of month): daily weather watch; full dress rehearsal; launch day - inflation, release, live tracking, chase and recovery; backup date one week later. Second half: all teams analyze flight data vs predictions; final technical reports & posters; datasets published on portal; lessons-learned register; public showcase/symposium.

**Flight Segment Focus:** Launch-day payload prep & final checks; in-flight health monitoring; post-recovery hardware inspection; prediction-vs-actual analysis (trajectory, thermal, power, link); science results written up.

**Ground Segment Focus:** Live mission operations: tracking, telemetry logging, dashboard ops, landing prediction updates for chase team; capture IQ recordings for BSAT-AIM-2026-005; post-flight link & packet-loss analysis; ML packet-recovery attempts; station handover documentation.

**Gate 5: Post-Flight Review (within 1 week of flight) • Gate 6: Final Presentations & Programme Review - end of Month 5**

# SYSTEM BUDGETS, INTERFACES & ORGANISATION

## SYSTEM BUDGETS (whole mission)

**Mass budget (target  $\leq 2$  kg):** Structure & insulation 400 g | Flight computer + power 350 g | Radio + antennas 150 g | Cameras + gimbal 300 g | Science payloads 500 g | Recovery aids + backup tracker 150 g | Margin 150 g

**Power budget (3-4 h mission):** Flight computer 0.5 W | Radio TX 1-2 W peak | Sensors 0.3 W | Cameras/edge-AI 3-4 W | Heaters (optional) 2 W | Total  $\approx 6-9$  W  $\rightarrow$  size battery  $\sim 2\times$  (30+ Wh usable at cold temperature)

**Telemetry budget:** LoRa  $\sim 433$  MHz: position packet every 5-10 s (priority 1), housekeeping every 30 s, science data every 60 s, image thumbnails opportunistic (priority lowest)

## KEY INTERFACES BETWEEN TEAMS

**BSAT-AVI-2026-001 Flight computer**  $\leftrightarrow$  **everyone:** Publishes the pinout, connector standard and data bus (I2C/SPI/UART allocation). All payload teams must submit an Interface Request Form by end of Month 2.

**BSAT-SWS-2026-004 Protocol**  $\leftrightarrow$  **BSAT-AVI-2026-003 radio**  $\leftrightarrow$  **BSAT-SWS-2026-002 ground:** One shared packet specification document is the single source of truth; version-controlled in Git; changes frozen at CDR (Month 3).

**BSAT-STR-2026-001 Structure**  $\leftrightarrow$  **all flying hardware:** Every flying item submits mass + dimensions + mounting needs by Month 2; BSAT-STR-2026-001 owns the master mass budget and can reject overweight items.

**BSAT-MEC-2026-003 Rotator**  $\leftrightarrow$  **BSAT-GND-2026-001 Yagi**  $\leftrightarrow$  **BSAT-GND-2026-004 software:** Mechanical mounting spec + rotctl protocol agreed in Month 2; integrated tracking test in Month 4.

**BSAT-MEC-2026-004 Sampler**  $\leftrightarrow$  **BSAT-PLC-2026-002 lab analysis:** The chemistry-payload team defines contamination requirements before mechanism design starts; joint seal-integrity test in Month 3.

**BSAT-MSN-2026-001/BSAT-DHA-2026-003/BSAT-AIM-2026-002 prediction stack:** Physics-based predictor (BSAT-MSN-2026-001) is the baseline; BSAT-DHA-2026-003 adds Monte Carlo uncertainty; BSAT-AIM-2026-002 attempts ML improvement. All three predict every test flight and compare.

## SUGGESTED PROGRAM ORGANIZATION

**Program structure:** 1 student Program Manager + 1 Chief Engineer (systems) + team leads per project. Weekly all-hands (30 min) + weekly subsystem meetings. Faculty advisor per subsystem team.

**Documentation:** Shared Git organization: one repo per team + one 'mission' repo for interface control document, budgets, packet spec, and reviews. All review gate slides archived.

**Review gates:** MRR → PDR → CDR → FRR → Post-Flight → Final. A team that misses a gate deliverable presents a recovery plan at the next all-hands.

**Safety:** RF safety (no TX near people during tests at full power), battery handling (LiPo bags), hydrogen safety if used (prefer helium), ladder/rooftop rules for antenna work, launch-day crowd control.

### ESTIMATED BUDGET (INDICATIVE)

**Flight hardware:** Balloon (×2) + gas + parachute: mid-range | Flight electronics & sensors: mid-range | Cameras: low-mid | Science payloads: varies by experiment

**Ground station:** SDR + LNA + filter: low | Antenna materials: low | Rotator motors & frame: mid | Computers: reuse existing lab machines

**Note:** Exact costs depend on country and suppliers - each team submits a bill of materials for approval at PDR (Month 2). Budget 15% contingency and buy 2 of every cheap critical part.

# ANNEXURE-1

## Project Selection & Team Registration Form

|                                        |  |
|----------------------------------------|--|
| <b>Project Code (from this Basket)</b> |  |
| <b>Project Title</b>                   |  |
| <b>Subsystem / Work Package</b>        |  |

| S.No | Team Member Name | Branch / Year | UID / Roll No | Signature |
|------|------------------|---------------|---------------|-----------|
| 1    |                  |               |               |           |
| 2    |                  |               |               |           |
| 3    |                  |               |               |           |
| 4    |                  |               |               |           |

|                                              |                                                                                                                                                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Team Lead (Name &amp; Contact)</b>        |                                                                                                                                                                                                                                   |
| <b>Faculty Mentor (Name &amp; Dept.)</b>     |                                                                                                                                                                                                                                   |
| <b>Prior Relevant Experience of the Team</b> |                                                                                                                                                                                                                                   |
| <b>Declaration</b>                           | We agree to follow the mission-level design constraints, review gates and safety rules stated in this Basket, and to hand over all hardware, software, data and documentation to the Programme Office at the end of the semester. |
| <b>Date &amp; Signatures</b>                 |                                                                                                                                                                                                                                   |

Submit the completed form to the Programme Office before the Mission Requirements Review (end of Month 1). One form per team; a student may be a member of only one team.

# ANNEXURE-2

## Proposal Format for Project Teams (max. 6 pages)

1. **Project Identification:** project code and title as given in this Basket; team composition and faculty mentor.
2. **Understanding of the Problem:** the team's own summary of the topic and its role in the BalloonSat mission (do not copy the Basket text).
3. **Technical Approach:** proposed design/method, candidate components or tools, and alternatives considered.
4. **Work Plan:** month-wise plan mapped to the five review gates (MRR, PDR, CDR, FRR, Flight & Final Review), with the deliverable committed at each gate.
5. **Interfaces:** mass, power, data and mechanical interface requests, referencing the linked project codes listed under "Linkages within the Mission".
6. **Test & Verification Plan:** how each Expected Deliverable will be tested, including environmental acceptance tests on the BSAT-STR-2026-004 rigs.
7. **Bill of Materials & Budget:** itemised cost estimate with 15% contingency; identify long-lead items to be ordered in Month 1.
8. **Risk Register:** top five technical risks with likelihood, impact and mitigation.
9. **References:** datasheets, papers, prior missions and community resources consulted.

Proposals are evaluated on technical soundness, feasibility within one semester, clarity of the verification plan and quality of the interface definition. Selected teams present at the Mission Requirements Review.



# OUR MENTORS

*Collaborating Today. Innovating Tomorrow*



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